FOOD RESOURCES

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→ The basic necessity for the survival of human beings is food. The food humans eat are composed of carbohydrates, proteins, vitamins etc.

The three main food sources are:→

(1) Croplands provide 76% of the total
(2) Rangelands that produce meat, milk accounts for 17% of total
(3) Fisheries supply remaining 7%.

⊛ World food Problems →

→ FAO (Food & agricultural organisation) of UN estimated that more than 800 million people lack access to the food. Most of these people live in rural areas of poorest developing countries. The 2 region of the world with the greatest food insecurity are South Asia and Sub-Saharan Africa.

→ The average human must consume 2600 kcal per day.

Average man → 3000 kcal per day
Average woman → 2200 kcal per day

• Undernourished people →

If a person consumes less than the required amount over an extended period his or her health & stamina

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decline. People who receive fewer calories than needed are undernourished

(*) <u>Malnourished</u> →

People can receive enough calories in their diets but still be <u>malnourished</u> because they are not receiving enough of specific, essential nutrients such as proteins, vitamin A, iodine, iron.

<u>For eg</u> → A person whose primary food is rice can obtain enough calories, but the diet of rice lacks sufficient amount of protein, minerals, vitamins.

Effects of malnutrition →
- Poor physical development
- Increased disease susceptibility
- children suffering from malnourishment does not grow normally

→ The 2 most common diseases of malnutrition are:
(a) Marasmus
(b) Kwashiorkor

- <u>Marasmus</u> :-

It is a progressive deterioration caused by a diet low in both total calories and protein. Symptoms are :→

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- Slow growth
- Extreme atrophy (wasting) of muscles

It is possible to reverse the effect of marasmus with an adequate diet.

- Kwashiorkor:

Malnutrition resulting from protein deficiency.

Symptoms are
- Dry, Brittle hair
- Stunted Growth
- Apathy
- Sometimes mental retardation.

Most typical feature is swelling of abdomen.

It is treated by restoring balanced diet

(*) Overnourished →

People who eat food in excess than required are overnourished. A person suffering from overnutrition has a diet high in fats, sugar, salt. Overnutrition results in obesity, high blood pressure, diabetes, heart disease. overnutrition is most common among people in highly developed nations such as united states.

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⊛ Impacts caused by Agriculture :→

It could be broadly classified into:
- on-site impacts
- Off-site impacts.

→ On site impacts include soil erosion and range land degradation.

- Soil erosion on agricultural lands take place through three major processes
  - Overland flow (Run off) - It is most visible and widespread.
  - Wind erosion - It occurs if strong wind blow at times when the soil surface is relatively dry
  - Streambank erosion
    → It is limited to fields that border streams.

- Rangeland Degradation

Overgrazing occurs when the number of animals on these lands exceeds carrying capacities. For eg → The dry lands around the Mediterranean Sea have been long overgrazed with resulting problems of devegetation, and the threat of desertification. Desertification is land degradation in dryland regions.

Overgrazing causes →
- Porosity of soil decreases
- Less fertile soil (less organic matter)

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Off site Impacts

- Water pollution from agricultural runoff
- Air pollution from blowing dust
- Dispersal of agricultural chemicals like fertilizers and pesticides.

(*) Types Of Agriculture :

- Industrialized or modern
- Subsistence agriculture

→ Most farmers in highly developed countries practice industrialized agriculture or high-input agriculture. It depends on large inputs of capital & energy in the form of fossil fuels, to run machinery, irrigate crops and produce fertilizers & pesticides. This type of agriculture produces high yield.

→ In the developing countries most farmers practice subsistence agriculture, the production of enough food to feed one's family with little left over to sell for hard times. Subsistence agriculture too requires large input of energy, but from humans & animals rather than fossil fuels.

Shifting cultivation is the form of subsistence agriculture in which short periods

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of cultivation are followed by longer periods of fallow land.
↳ (land left uncultivated).

→ Slash and burn agriculture is one of the distinct type of shifting cultivation that involves clearing small patches of tropical forests to plant crops. They move from one area to another because soil looses its productivity when they are cultivated. This agriculture is land-intensive.

- Nomadic herding is another land-intensive subsistence agriculture
- Intercropping is another form of intensive subsistence agriculture that involves growing a variety of plants simultaneously on same field.

Green Revolution →

In the middle of 20th century, serious food shortages occured in developing countries like India. It was recognized that additional food supplies were needed to feed growing population. The development of high yielding varieties of wheat & rice gave the chance to provide the people with adequate supplies of food but for producing them intensive cultivating methods, use of pesticides & fertilizers were needed. →

using modern cultivation methods and the high yielding varieties of certain staple crops to produce more food per acre of cropland is known as green revolution

→ The 2 most imp. problems associated with higher crop production are
- High energy costs
- Serious environmental problems caused by the use of fertilizers & pesticides.

(*) Effects of Modern / Industrialized agriculture →

The practices of industrialized agriculture have resulted in several environmental problems.

- The agricultural use of fossil fuels produces air pollution.
- Agricultural chemicals such as fertilizers and pesticides cause water pollution that harm fisheries etc. The agricultural practices are one of the major cause of surface water pollution in India.

* Fertilizers & Pesticides

Fertilizer use was increased upto 4 times btw ~~19610~~ 1960 to 1990. One of the reason was that for the development of high yielding plant the

Fertilizer Problems →
- Health Hazards (Blue Baby Syndrome)
- Soil Damage
- Fertilizer Burn
- Fertilizer Run-off

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large inputs of fertilizers were needed.

→ The three most important nutrients required by plants are nitrogen, phosphorous & potassium. Nitrogen is ultimately derived from the atmosphere, but it is made available to plants by nitrogen-fixing bacteria. It is the nutrient that is most often dificient and is most widely applied to crops. Natural gas is an important raw material in the manufacture of nitrogen fertilizer. The most commonly used forms are ammonia & urea.

Phosphorous is present in small quantities in soil

→ Phosphorous is applied as superphosphate or as phosphoric acid, which are manufactured from phosphate rock deposits.

→ ~~Potassium~~ plants demand potassium in large quantities and in many areas potassium fertilization is important.

→ Organic fertilizers (manure) is the most important source of nutrient like nitrogen. They help maintain good soil structure and they keeps organic matter content high. Manure is difficult to apply as compared to other fertilizers.

Inorganic fertilizers have replaced organic fertilizer (Manure) because
- Manure is low in nutrient content

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→ Pesticides is chemical agents used to control organisms harmful to plants. Pesticides include insecticides, rodenticides, fungicides and others. Herbicides are used to control weeds

→ The first widely used insecticides were organochlorines such at DDT, aldrin. Pesticides & Herbicides are used on corn & soybeans.

→ Some of the well-known health effects of pesticide exposure include acute poisoning, cancer, neurological effects, reproductive & developmental harm. The major cause of concern are bio-accumulation of pesticides.

⊛ Indian Scenario

In the last 50 years the use of fertilizers in India has grown nearly 150 times. The consumption of insecticide has been increased more than 100% from 1971 - 1995.

⊛ Salinity & Waterlogging :→

Salinity is when the concentration of soluble salts in the soil increases. This is mostly due to low rainfall, poor drainage and high temperatures, when

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water evaporates quickly leaving behind the highly concentrated salts.

Salinity can be prevented by :→
- Improving the drainage
- Reclamation of salinated lands by leaching with plenty of water.

→ Waterlogging occurs when the soil surface area becomes saturated and soil pores are full of water.

~~Watt~~, Water logging occur due to
- Periods of heavy rain
- Poor irrigation management
- Poor drainage
- Rising water table

Water table could also rise due to overwatering with irrigation.

→ Effects of water logging are :→

1) Water logged soil pores have no oxygen. Plants need oxygen to breathe & grow
2) Vegetation can turn yellow, growth is stunted
3) Trees & plants can die
~~4)~~

→ Prevention:
1) Management of drainage lines for efficient water flow

2) Management of surface water flow to avoid ~~water~~ surface ponding.
3) Increase deep rooting vegetation for greater utilisation of water from the soil.

In India waterlogging is prevalent in the coastal areas of Mumbai, Kerala, Andaman & Nicobar Islands.

(*) Sustainable Agriculture →

Negative effects of agriculture are being felt [soil degradation, agricultural pollution]. But the need of the hour is to increase food production.
Farm practices that seek to balance production & preservation are known as 'sustainable agriculture'.

Intensive soil conservation measures and minimal use of pesticides & fertilizers constitute sustainable agriculture. The farmers producing grain, meat, dairy products using sustainable methods is increasing and this trend is likely to continue as long as the cost of chemical input remain high.

→ One key feature of the drive toward sustainable agriculture is the reduced use of pesticides. Though it has been a boon to ~~agr~~ modern agriculture but it also has harmful side effects.

- Health hazards to agricultural workers using pesticides
- health effects on general public through contamination of food, water etc.

Foreg → Nonpersistent pesticides which are used nowadays do not accumulate in high concentration in the environment. But they are highly toxic at the time when they are applied, hence those in contact are at risk. (Biomagnification)

End